



AMITY UNIVERSITY
UTTAR PRADESH

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

NTCC Project Diary

B. Tech (Computer Science and Engineering)

Group No: 148

Project Title: Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches

Area: Artificial Intelligence and Machine Learning (Computer Vision)

Academic Session: 2026-27

Project Guide: Dr. Richa Gupta

Details of Project Team:

Programme:- B.TECH CSE		Year/Semester:- 4 th Year /7 th Semester	
S. No.	Enrollment No.	Name	Signature
1.	A2305223472	Rohit Yadav	
2.	A2305223568	Vishal	
3.	A2305223449	Hardik Mehlawat	

Project Diary:

Record of work carried out on the minor project between 20 July 2026 and 27 August 2026, in steps of two to three days. Each entry corresponds to material held in the project's working repository.

S. No.	Duration	Work carried out	Signature of Guide
1.	20/07/2026 - 22/07/2026	Project scope fixed as detection of AI-generated still images, that is, binary classification of an image as camera-captured or synthesised by a generative model. Video deepfake detection, audio detection and any deployed product were recorded as out of scope. Working repository set up for the semester with separate folders for section drafts, tables, figures, references and progress reports. Candidate titles drafted for the guide's approval.	

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2.	23/07/2026 - 26/07/2026	Literature search protocol written and recorded so that it can be reported in the paper itself: six databases (IEEE Xplore, ScienceDirect, SpringerLink, the ACM Digital Library, arXiv and Google Scholar), eight query seeds covering GAN, diffusion and dataset-specific terms, inclusion limited to still-image work from 2019 to 2026 reporting quantitative results, and exclusion of video-only, audio-only, non-English and inaccessible work. Section skeleton for the review paper fixed at eight sections with a word budget for each.	
3.	27/07/2026 - 29/07/2026	Project title finalised as “Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches” and adopted as the official minor project topic. Official Weekly Progress Report proforma received from the department and replicated as a one-page template for the semester's reports. Database searches run against the eight query seeds.	
4.	30/07/2026 - 02/08/2026	Search results consolidated: 101 records identified, 26 duplicates removed, leaving 75 unique candidate papers recorded with venue and DOI or arXiv identifier. Title-and-abstract screening completed against the inclusion and exclusion criteria, shortlisting 48 primary studies for full-text reading, 11 secondary studies for selective use and 15 background references on generators, backbones and explanation methods. Stage counts recorded for the search-flow figure.	
5.	03/08/2026 - 05/08/2026	Full-text reading of the first half of the 48 primary studies. For each study read, the eight-column extraction table was populated: detection approach, model or architecture, dataset used, generators covered, reported accuracy or AUC, whether a cross-generator test is reported, and the limitation stated by the authors themselves. Cells that could not be confirmed from a paper's text were marked for verification rather than filled with an estimate.	
6.	06/08/2026 - 09/08/2026	Full-text reading of the remaining primary studies completed and the extraction table populated for all 48. Citation identifiers checked during reading: three DOIs confirmed, one missing DOI located, one preprint replaced by its published journal version, and one preprint excluded for failing the well-cited criterion of the search protocol.	
7.	10/08/2026 - 12/08/2026	Taxonomy of detection approaches finalised and all 48 studies assigned to its five families: 14 handcrafted and frequency-domain, 19 CNN-based and transfer learning, 2 transformer and attention-based, 8 multimodal and foundation-model-based, and 4 explainability. Section 1 (Introduction) and Section 2 (Background) drafted. Reference infrastructure completed: a BibTeX file built for all 74 included references, every identifier opened and checked against its arXiv, Crossref or proceedings record, and one publication-year error corrected.	
8.	13/08/2026 - 14/08/2026	Sections 3 to 8 drafted: the five taxonomy subsections, the datasets section with an 11-dataset comparison table, the metrics section with eight numbered equations, the comparative analysis built on the master table of all 48 studies, the gap analysis, the conclusion and the abstract. The master table was checked against the extraction records by script and all 48 rows matched. Two figures produced: the search-flow diagram and the metrics overview panel.	

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9.	15/08/2026 - 17/08/2026	Full draft assembled into a formatted report document with both comparison tables inserted where the text first refers to them and both figures embedded, then exported to a text-searchable PDF and submitted for the departmental plagiarism check. On the same day the project guide instructed the team to begin preparing the implementation code ahead of Semester VIII, so a second work track was opened alongside the paper.	
10.	18/08/2026 - 19/08/2026	Implementation plan written, covering hardware, dataset handling, the experiment matrix, the training defaults with a one-line justification for each, and the schema of the results CSV. Local Python environment built and version-pinned. CIFAKE downloaded and its train, validation and test split generated with seed 42 and committed to the repository: 90,000 training, 10,000 validation and 20,000 test images, balanced between the real and generated classes.	
11.	20/08/2026 - 21/08/2026	Six pipeline modules implemented — data loading, model construction, training, evaluation, Grad-CAM and results merging — together with the ten experiment configuration files of the matrix. An end-to-end smoke run was completed on the laptop CPU, producing a results row, a checkpoint that reloads and a Grad-CAM overlay. A code review was carried out and its findings fixed before any GPU time was spent.	
12.	22/08/2026 - 23/08/2026	Pipeline verified on the Kaggle GPU: the smoke run reproduced the CPU result exactly, which is evidence that the protocol is reproducible across machines, and per-architecture timings were measured so that training sessions could be sized against the weekly GPU allowance. First three training sessions run, completing ResNet50, DenseNet121 and EfficientNet-B0 in both modes and the Vision Transformer in feature-extraction mode.	
13.	24/08/2026 - 25/08/2026	VGG19 completed in both modes. Its two results were recorded as the clearest single finding so far: last of the five backbones under feature extraction at 90.55% test accuracy and second under fine-tuning at 97.91%, the largest gap between the two modes in the matrix. The publication plan was revised with the guide's approval to a single merged research paper covering both semesters' work, with the two Amity project reports left unchanged.	
14.	26/08/2026 - 27/08/2026	Vision Transformer fine-tuning completed, closing the experiment matrix at ten runs out of ten, all on the same GPU type with the same seed and split file, for 19.89 hours of measured GPU time. Results merged into the results CSV and the three runs that stopped at the epoch ceiling flagged as lower bounds, with reruns prepared. Progress report and project diary prepared for the First Progress Review Presentation on 27 August 2026.	

Signature(s) of project team:

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Signature of project guide

Dr. Richa Gupta

Date: